The empirical recurrence for OEIS A264580, recovered from its tail

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Abstract

OEIS A264580 records "Empirical recurrence of order 84" and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 4096$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the whole sequence, so the recurrence holds from $n = 85$ — every index at which it can be written down. Its last coefficient is $c_{84} = -1 \neq 0$, so no further tail satisfies anything shorter; any order-84 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry's recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A264580 is "Number of (n+1)X(5+1) arrays of permutations of 0..n*6+5 with each element having directed index change 1,0 0,-1 1,2 or -1,1.". It has offset 1 and begins

$$1, 16, 97, 553, 3668, 24370, 145761, 905291, \ \ldots$$

The entry states:

> Empirical recurrence of order 84 (see link above)

As of the "Last modified" line on the live entry (Nov 18 2015 20:32:34, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry's condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u, and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) \ = \ \iota^{\top} M^{\,j(n)} \tau$$

for a linear reindexing j fixed by the entry's shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 4096$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 4096$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S'+1$ consecutive values of a. $\square$

3 Why the question has to be asked of a tail

Here the stated order is already the minimal order of the sequence itself, so no tail has to be taken.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 8192$ exact terms from $n = 1$ on, it returns order 84 — the order the entry states — with integer coefficients:

$$a(n) \;=\; \sum_{i=1}^{84} c_i\, a(n-i),$$

c_1	6	c_{22}	−4960133698	c_{43}	728581835742	c_{64}	−462301123
c_2	−5	c_{23}	6587003892	c_{44}	−411838528825	c_{65}	189712325
c_3	97	c_{24}	−5121154437	c_{45}	79988591222	c_{66}	−82217324
c_4	−380	c_{25}	22911258255	c_{46}	−529205762573	c_{67}	85811480
c_5	340	c_{26}	−26932848972	c_{47}	268913246513	c_{68}	−39414548
c_6	−3077	c_{27}	18505580837	c_{48}	−32722790430	c_{69}	18893774
c_7	13883	c_{28}	−78611226969	c_{49}	293942811160	c_{70}	−11482184
c_8	−16496	c_{29}	81450029297	c_{50}	−132857489758	c_{71}	5247226
c_9	56474	c_{30}	−48482316794	c_{51}	5478629108	c_{72}	−2360552
c_{10}	−358771	c_{31}	204637761665	c_{52}	−123687605208	c_{73}	944049
c_{11}	565460	c_{32}	−186370753126	c_{53}	48480037023	c_{74}	−414879
c_{12}	−896297	c_{33}	94283167076	c_{54}	2596804599	c_{75}	169006
c_{13}	6663553	c_{34}	−410254526583	c_{55}	39441224160	c_{76}	−41114
c_{14}	−11410312	c_{35}	328516069398	c_{56}	−13064087299	c_{77}	18845
c_{15}	12624749	c_{36}	−138315543659	c_{57}	−1866464151	c_{78}	−6927
c_{16}	−86341755	c_{37}	638711060509	c_{58}	−9900268525	c_{79}	752
c_{17}	141615239	c_{38}	−451772450602	c_{59}	2914822953	c_{80}	−493
c_{18}	−134714856	c_{39}	154143851539	c_{60}	289745226	c_{81}	149
c_{19}	778083870	c_{40}	−774096258818	c_{61}	2179597617	c_{82}	−2
c_{20}	−1158994808	c_{41}	487558477449	c_{62}	−711700564	c_{83}	5
c_{21}	999718631	c_{42}	−129730703741	c_{63}	141979054	c_{84}	−1

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 85$, and it is the recurrence OEIS A264580 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 1$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{84} - c_1 x^{84-1} - \cdots - c_{84}$. Its constant term is $-c_{84} = 1$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 84 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 84, so $\deg q = 84$, and it is monic. It holds from some index t on. From $\max(t, 1)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 84, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 20 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry's English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 84, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry's.

Third, the recovered recurrence was evaluated directly on the entry's own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -1.

Remark 4. The conjecture's own statement was never read. The entry pins it down to the family of order-84 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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